



Booked Full, Sat Empty

Validating the Timetabling Office's Occupancy Forecast Against a Term of Clicker-Recorded Attendance

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The problem

Timetabling caps or grows a subject's enrolment against a pre-term occupancy forecast. Whether that forecast still describes the room once the term is under way is rarely checked against what the room actually does.

Research questions

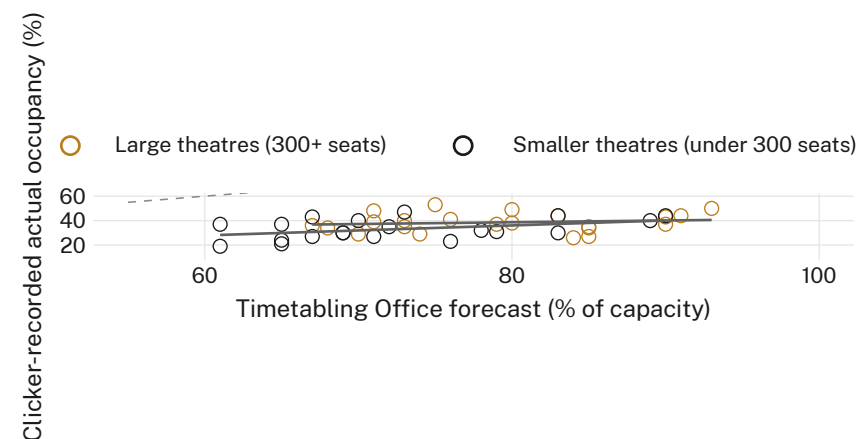
- Does the forecast track clicker-recorded attendance across a term?
- Does the forecast-to-actual gap narrow as the term progresses?
- Does the gap differ between large and small lecture theatres?

Study design

- n = 214 timetabled sessions · 9 lecture theatres · ceiling-mounted infrared clickers, 5-min sampling · 13-week term, Semester 1 2026
- session forecasts drawn from the Timetabling Office's own pre-term capacity model, held fixed throughout (no mid-term recalibration)
- theatres classified 300+ / under-300 seats before data collection

Findings

Forecast occupancy correlates weakly with clicker-recorded attendance ($r = 0.38$, $n = 214$). The forecast exceeds recorded attendance in 189 of 214 sessions (88%), a margin that does not narrow between week 1 and week 13.



Forecast occupancy exceeded clicker-recorded attendance in 88% of the term's 214 sessions; the median overshoot widened from 21 points in week 1 to 26 points in week 13, Semester 1 2026.

Implications & next steps

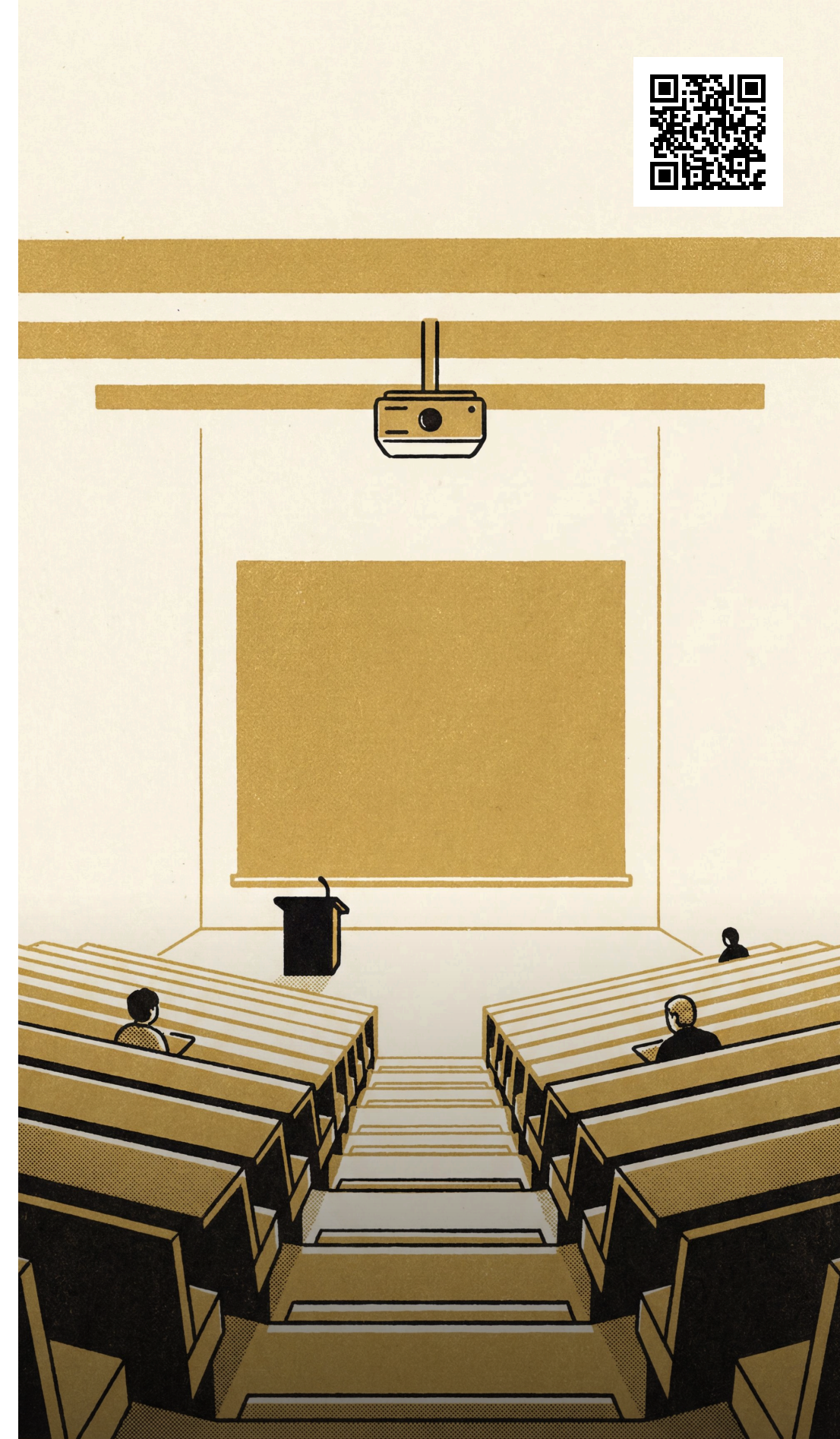
The forecast tracks the timetable, not the room — a model built on historical enrolment outlives the cap it once justified. Next: recalibrate mid-term against the clicker feed.

Selected references

1. Lewis, R. (2007). Survey of metaheuristic-based techniques for University Timetabling problems. *OR Spectrum*, 30(1), 167–190. doi:10.1007/s00291-007-0097-0
2. Müller, T. et al. (2025). Real-world university course timetabling at ITC 2019. *Journal of Scheduling*, 28(2), 247–267. doi:10.1007/s10951-023-00801-w
3. Violatto, G. et al. (2019). Anomalous occupancy sensor behavior detection in connected indoor lighting systems. *IEEE WF-IoT 2019*, 335–339. doi:10.1109/wf-iot.2019.8767217
4. Heidarifar, H., & Shahbakhti, M. (2023). Occupancy detection in a residential building using sensor fusion and ML. *Progress in Canadian Mechanical Engineering*, 6. doi:10.17118/11143/20841
5. Mahgoub, Y. O. M., & Khalil, R. (2012). Energy utilization and human behavior in higher education facilities. *Qatar Foundation Annual Research Forum*, 2012(1). doi:10.5339/qfarf.2012.ahp18

Clicker counts aggregated to session level only; no student identifiers retained. Sensor placement audited termly by Facilities.

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“The forecast says three-quarters full.
The clickers say a third.”

214 timetabled sessions, nine lecture theatres, one 13-week term.